

Probabilistic Robotics Course

Gaussian Distribution

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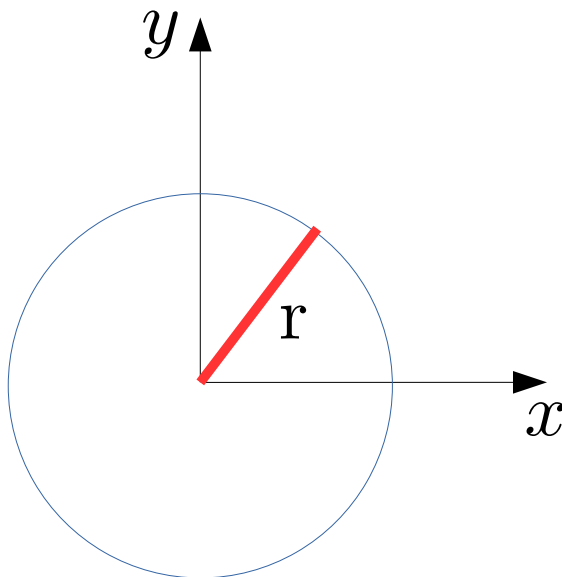
Outline

- Drawing Ellipses
- Parametrizations
- Drawing Gaussians
- Classical Parametrization
 - Marginalization
 - Conditioning
 - Chain Rule
 - Affine Functions
 - Quasi-Affine Functions

Circles

A circle looks like that:

$$x^2 + y^2 = r^2$$

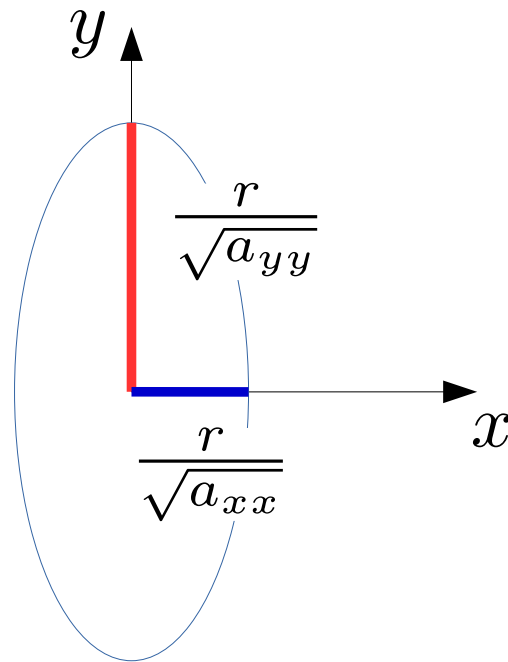


It is a slice of a paraboloid.

Scaled Circles

A *scaled* circle like that:

$$a_{xx}x^2 + a_{yy}y^2 = r^2$$



Slanted Circles

A *slanted scaled* circle like that:

$$a_{xx}x^2 + a_{xy}xy + a_{yy}y^2 = r^2$$

can be rewritten as:

$$\begin{pmatrix} x & y \end{pmatrix} \underbrace{\begin{pmatrix} a_{xx} & \frac{a_{xy}}{2} \\ \frac{a_{xy}}{2} & a_{yy} \end{pmatrix}}_{\mathbf{A}} \begin{pmatrix} x \\ y \end{pmatrix} = r^2$$

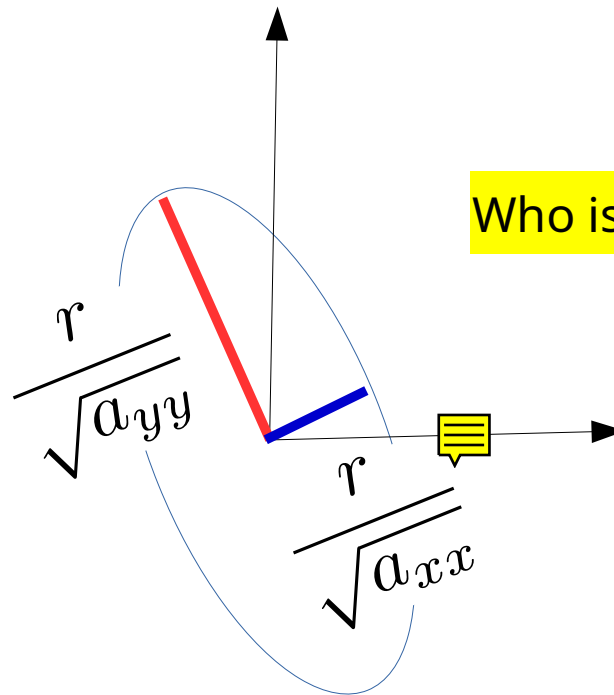
The **matrix** $\mathbf{A}$ admits an *eigenvalue decomposition*

$$\mathbf{A} = \mathbf{R}^T \begin{pmatrix} \lambda_1 & \\ & \lambda_2 \end{pmatrix} \mathbf{R}$$

Slanted Circles (cont.)

All this because of a_{xy}

The off diagonal components “rotate” the ellipsoid.

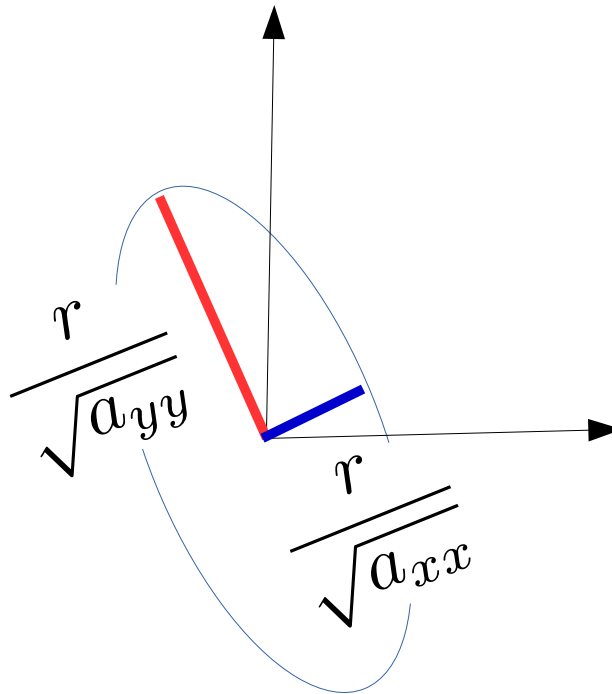


Who is guilty for the rotation?

Breaking News

Ellipses can also be translated:

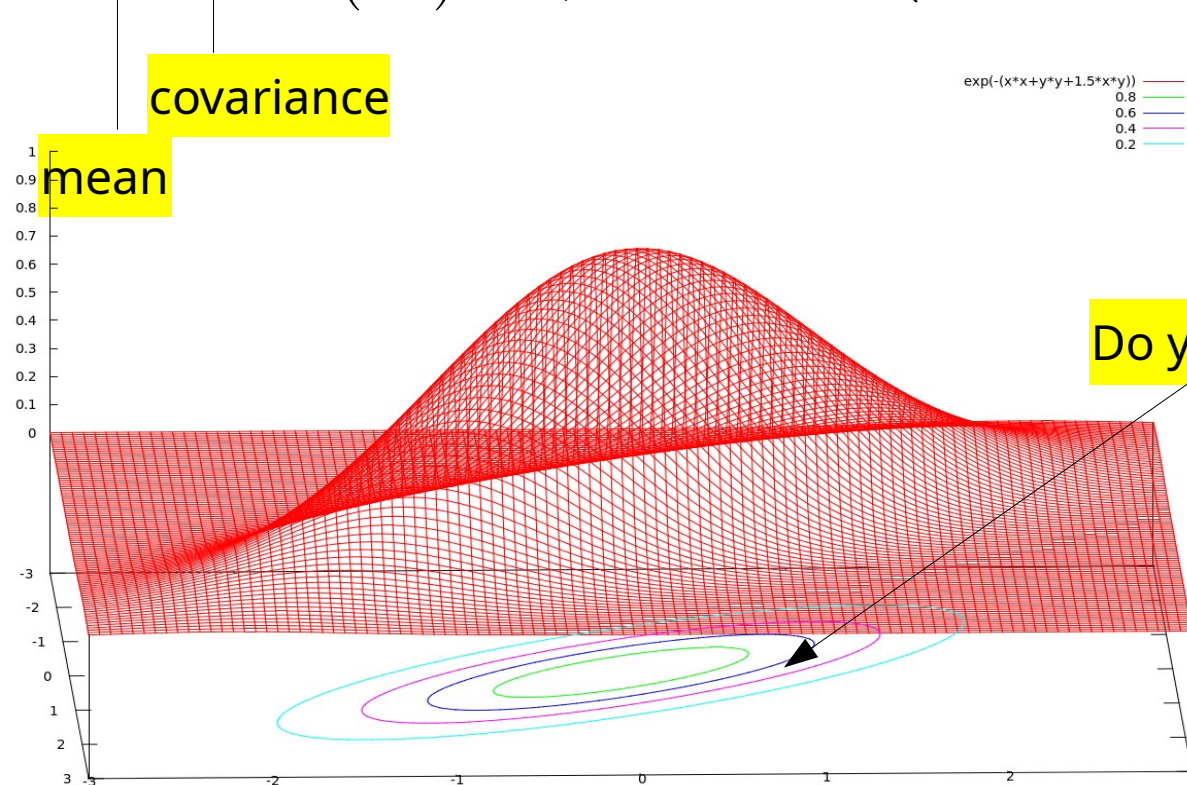
$$\mathbf{A} = \left[\mathbf{R} \begin{pmatrix} x - x_c \\ y - y_c \end{pmatrix} \right]^T \begin{pmatrix} \lambda_1 & \\ & \lambda_2 \end{pmatrix} \left[\mathbf{R} \begin{pmatrix} x - x_c \\ y - y_c \end{pmatrix} \right] = r^2$$



Gaussian

The PDF of a Gaussian distribution has the following form:

$$p(\mathbf{x}) = \mathcal{N}(\mathbf{x}; \mu, \Sigma) = \frac{1}{(2\pi)^{n/2} \sqrt{\det \Sigma}} \exp \left(-\frac{1}{2} (\mathbf{x} - \mu)^T \Sigma^{-1} (\mathbf{x} - \mu) \right)$$



covariance

mean

Do you see the ellipse???

Why Gaussians are cool

Gaussian distributions are closed under:

- sum
- affine transformation ($Ax+b$)
- conditioning
- marginalization

This means that in order to implement the above operations, one only needs to compute the **parameters** of the result, from the parameters of the input

Moment Parametrization

The one seen in the previous slide is known as *moment parameterization*.

The parameters can be calculated from a (large) set of samples as:

$$\mu = \frac{1}{N} \sum \mathbf{x}^{(i)}$$

$$\Sigma = \frac{1}{N-1} \sum (\mathbf{x}^{(i)} - \mu)(\mathbf{x}^{(i)} - \mu)^T$$

Moment Parametrization

The parameters are defined as the 1st and 2nd order moments of the distribution:

$$\mu = \int_{\Omega} \mathbf{x} p(\mathbf{x}) d\mathbf{x} = \mathbb{E}[\mathbf{x}]$$

$$\Sigma = \int_{\Omega} (\mathbf{x} - \mu)(\mathbf{x} - \mu)^T p(\mathbf{x}) d\mathbf{x} = \mathbb{E}[(\mathbf{x} - \mu)(\mathbf{x} - \mu)^T]$$

Canonical Parametrization

Another parametrization is the so called *canonical*, useful for conditioning:

$$\text{information matrix } \mathbf{\Omega} = \mathbf{\Sigma}^{-1} \quad \text{☰}$$

$$\text{information vector } \boldsymbol{\nu} = \mathbf{\Omega} \boldsymbol{\mu}$$

$$\mathcal{N}^{-1}(\mathbf{x}; \boldsymbol{\nu}, \mathbf{\Omega}) = \frac{\exp\left(\frac{1}{2}\boldsymbol{\nu}^T \mathbf{\Omega}^{-1} \boldsymbol{\nu}\right) \sqrt{\det \mathbf{\Omega}}}{(2\pi)^{n/2}} \exp\left(-\frac{1}{2}\mathbf{x}^T \mathbf{\Omega} \mathbf{x} + \mathbf{x}^T \boldsymbol{\nu}\right)$$

Partitioned Gaussian Densities

The space can be split in two subspaces.

The density is over a joint distribution:

$$\mathbf{x} = \begin{pmatrix} \mathbf{x}_a \\ \mathbf{x}_b \end{pmatrix} \quad \mu = \begin{pmatrix} \mu_a \\ \mu_b \end{pmatrix} \quad \nu = \begin{pmatrix} \nu_a \\ \nu_b \end{pmatrix}$$
$$\Sigma = \begin{pmatrix} \Sigma_{aa} & \Sigma_{ab} \\ \Sigma_{ba} & \Sigma_{bb} \end{pmatrix} \quad \Omega = \begin{pmatrix} \Omega_{aa} & \Omega_{ab} \\ \Omega_{ba} & \Omega_{bb} \end{pmatrix}$$

Affine Transformation

Let $\mathbf{X}_a$ be a Gaussian random variable such that:

$$\mathbf{x}_a \sim \mathcal{N}(\mathbf{x}_a; \mu_a, \Sigma_a)$$

Let

$$\mathbf{x}_b = \mathbf{f}(\mathbf{x}_a) = \mathbf{A}\mathbf{x}_a + \mathbf{c}$$

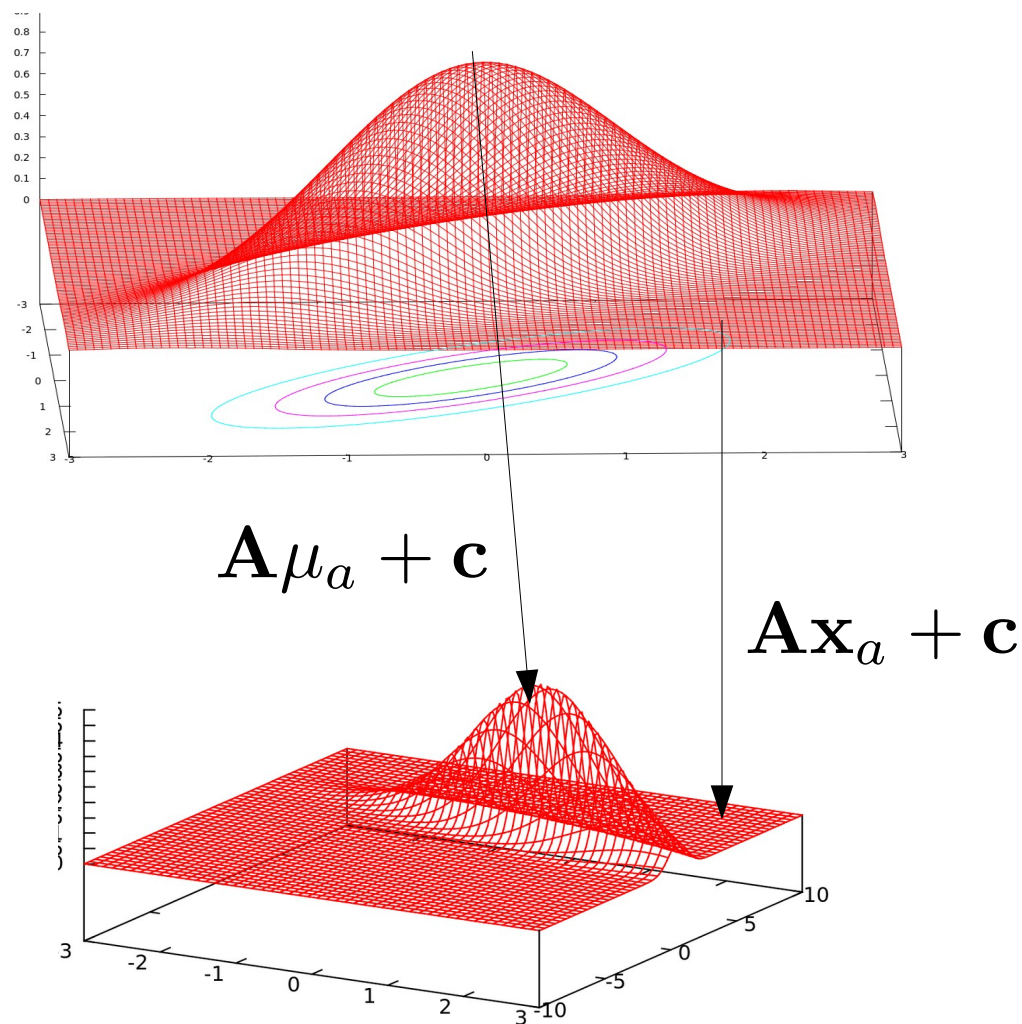
be an affine transformation of $\mathbf{X}_a$.

$\mathbf{X}_b$ is Gaussian:

$$p(\mathbf{x}_b) = \mathcal{N}(\mathbf{x}_b; \mu_b, \Sigma_b)$$

The parameters are:

$$\mu_b = \mathbf{A}\mu_a + \mathbf{c} \quad \Sigma_b = \mathbf{A}\Sigma_a\mathbf{A}^T$$



Taylor Expansion

For non-linear transformations, we can approximate the function around a *linearization point* $\mathbf{x}_0$

$$\begin{aligned} \mathbf{f}(\mathbf{x}) &\simeq \mathbf{f}(\mathbf{x}_0) + \underbrace{\frac{\partial \mathbf{f}(\mathbf{x})}{\partial \mathbf{x}}}_{\mathbf{A}} (\mathbf{x} - \mathbf{x}_0) \\ &= \mathbf{A}\mathbf{x} + \underbrace{\mathbf{f}(\mathbf{x}_0) - \mathbf{A}\mathbf{x}_0}_{\mathbf{b}} \end{aligned}$$

- This reduces the transformation to an affine transform
- The approximation holds only around a linearization point.
- The farther $\mathbf{f}$ is from being linear, the worse the approximation

Marginalization

Let $\mathbf{x}^T = (\mathbf{x}_a^T \mathbf{x}_b^T)$ be a Gaussian random variable such that

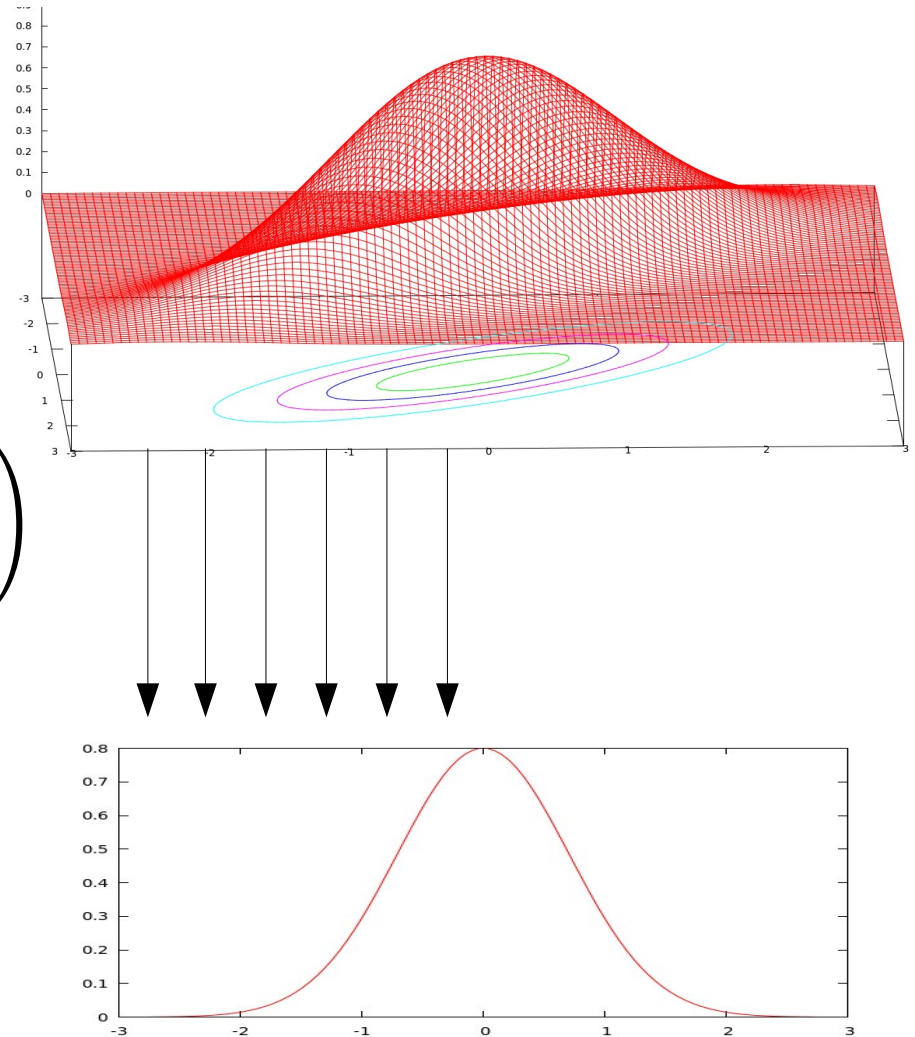
$$\mathbf{x} \sim \mathcal{N}(\mathbf{x}; \mu, \Sigma)$$

$$\Sigma = \begin{pmatrix} \Sigma_{aa} & \Sigma_{ab} \\ \Sigma_{ba} & \Sigma_{bb} \end{pmatrix}$$

$$p(\mathbf{x}_a) = \int_{\mathbf{x}_b} p(\mathbf{x}_a, \mathbf{x}_b) d\mathbf{x}_b$$

The marginal is Gaussian with parameters

$$p(\mathbf{x}_a) = \mathcal{N}(\mathbf{x}_a; \mu_a, \Sigma_{aa})$$



Conditioning (1)

Let $\mathbf{x}^T = (\mathbf{x}_a^T \ \mathbf{x}_b^T)$

Gaussian random variable such that

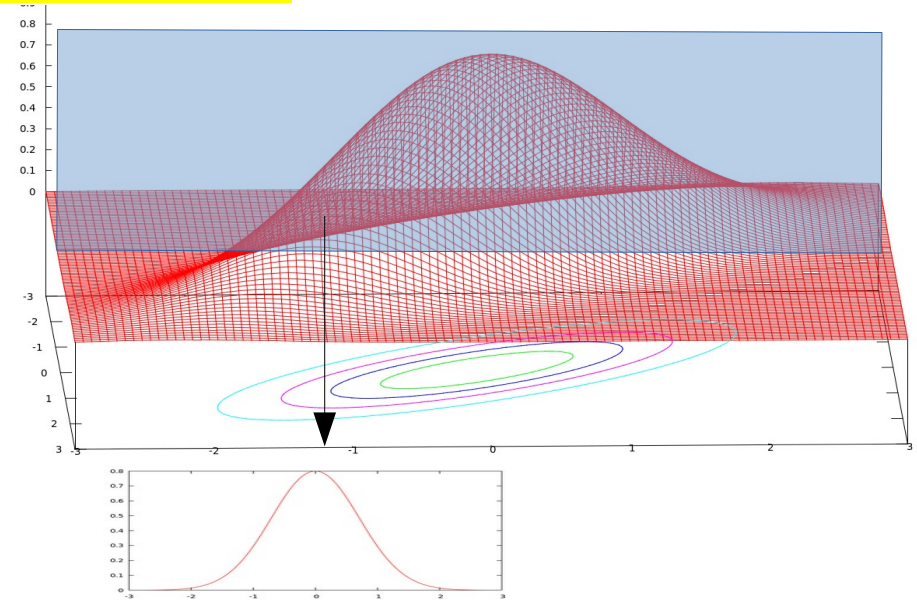
$$\mathbf{x} \sim \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma})$$

The conditional

$$p(\mathbf{x}_a \mid \mathbf{x}_b) = \frac{p(\mathbf{x}_a, \mathbf{x}_b)}{\int_{\mathbf{x}_a} p(\mathbf{x}_a, \mathbf{x}_b) d\mathbf{x}_a}$$

is Gaussian with parameters $\equiv$

$$p(\mathbf{x}_a) = \mathcal{N}(\mathbf{x}_a; \mu_{a|b}, \boldsymbol{\Sigma}_{a|b})$$



$$\mu_{a|b} = \mu_a + \boldsymbol{\Sigma}_{ab} \boldsymbol{\Sigma}_{bb}^{-1} (\mathbf{x}_b - \mu_b)$$

$$\boldsymbol{\Sigma}_{a|b} = \boldsymbol{\Sigma}_{aa} - \boldsymbol{\Sigma}_{ab} \boldsymbol{\Sigma}_{bb}^{-1} \boldsymbol{\Sigma}_{ba}$$

Conditioning (2)

Let $\mathbf{x}^T = (\mathbf{x}_a^T \ \mathbf{x}_b^T)$ be a Gaussian random variable such that

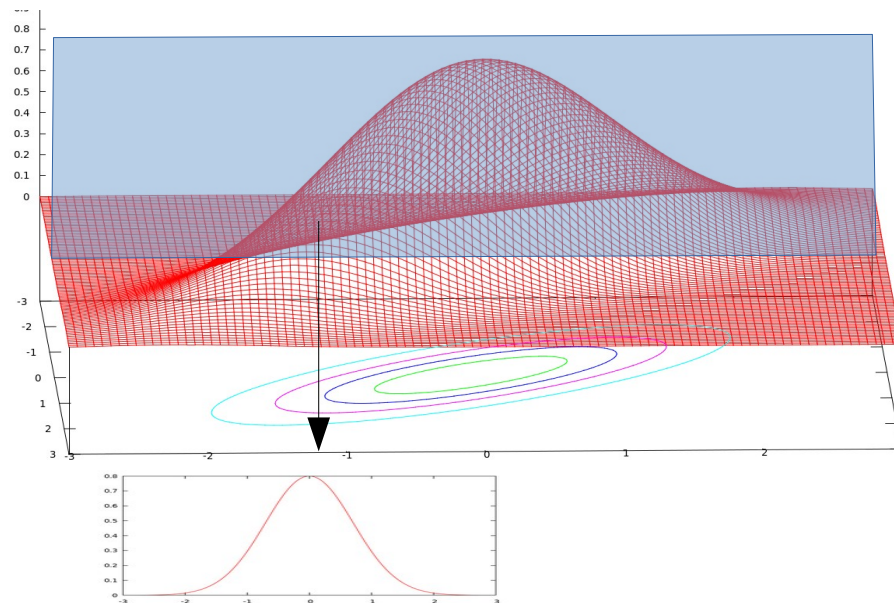
$$\mathbf{x} \sim \mathcal{N}^{-1}(\mathbf{x}; \boldsymbol{\nu}, \boldsymbol{\Omega})$$

The conditional

$$p(\mathbf{x}_a \mid \mathbf{x}_b)$$

is Gaussian with parameters

$$p(\mathbf{x}_a) = \mathcal{N}^{-1}(\mathbf{x}_a; \boldsymbol{\nu}_{a|b}, \boldsymbol{\Omega}_{a|b})$$



$$\boldsymbol{\nu}_{a|b} = \boldsymbol{\nu}_a - \boldsymbol{\Omega}_{ab}\mathbf{x}_b$$

$$\boldsymbol{\Omega}_{a|b} = \boldsymbol{\Omega}_{aa}$$

Chain Rule (1)

We know

$$p(\mathbf{x}_a) = \mathcal{N}(\mathbf{x}_a; \mu_a, \Sigma_a).$$

$$p(\mathbf{x}_b | \mathbf{x}_a) = \mathcal{N}(\mathbf{x}_b; \underbrace{\mathbf{A}\mathbf{x}_a + \mathbf{c}}_{\mu_{b|a}}, \Sigma_{b|a})$$

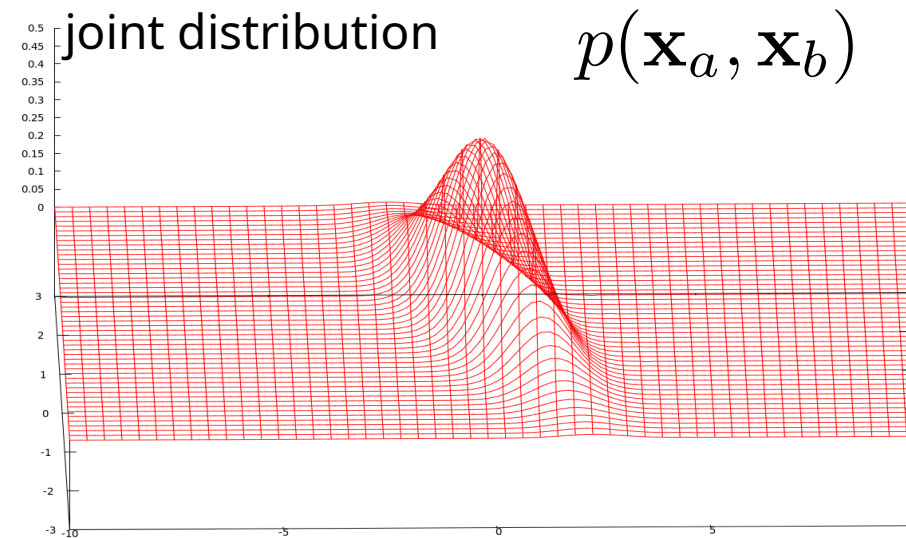
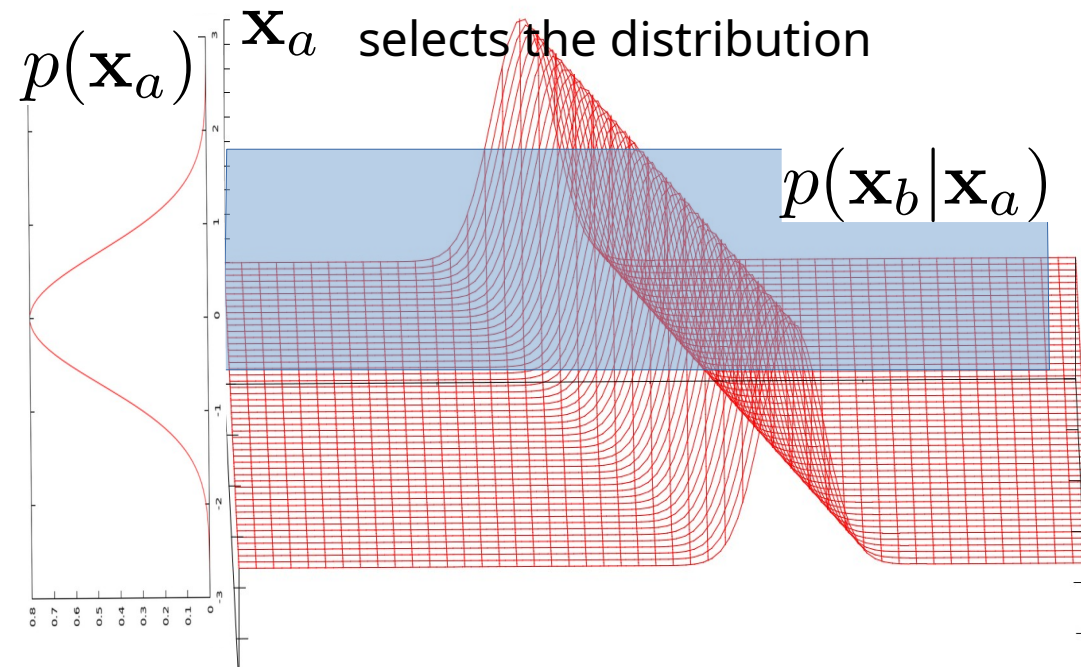
We want to compute

$$p(\mathbf{x}_a, \mathbf{x}_b) = \mathcal{N}(\mathbf{x}_{a,b}; \mu_{a,b}, \Sigma_{a,b})$$

The parameters are

$$\mu_{a,b} = \begin{pmatrix} \mu_a \\ \mu_b \end{pmatrix} = \begin{pmatrix} \mu_a \\ \mathbf{A}\mu_a + \mathbf{c} \end{pmatrix}$$

$$\Sigma_{a,b} = \begin{pmatrix} \Sigma_a & \Sigma_a \mathbf{A}^T \\ \mathbf{A}\Sigma_a & \Sigma_{b|a} + \mathbf{A}\Sigma_a \mathbf{A}^T \end{pmatrix}$$



Chain Rule (2)

We know

$$p(\mathbf{x}_a) = \mathcal{N}(\mathbf{x}_a; \mu_a, \Sigma_a).$$

$$p(\mathbf{x}_b | \mathbf{x}_a) = \mathcal{N}(\mathbf{x}_b; \mathbf{A}\mathbf{x}_a + \mathbf{c}, \Sigma_{b|a})$$

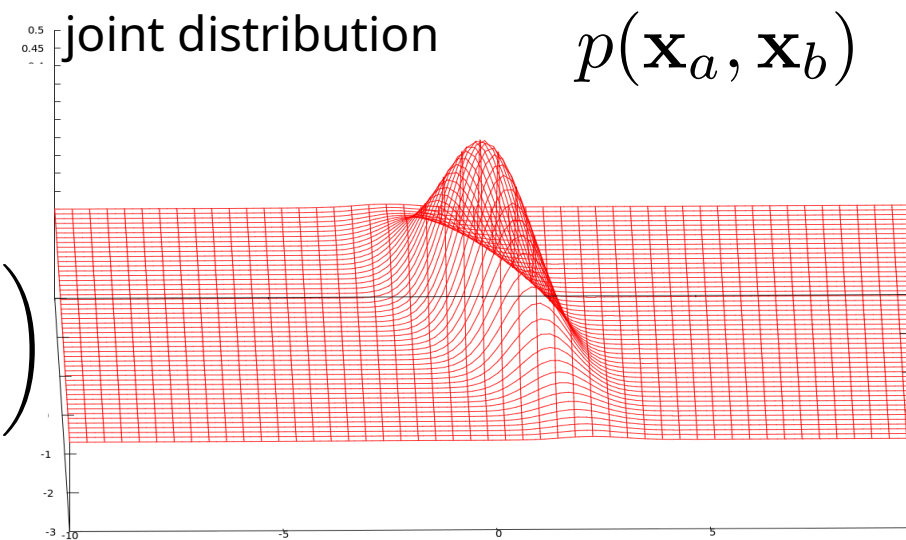
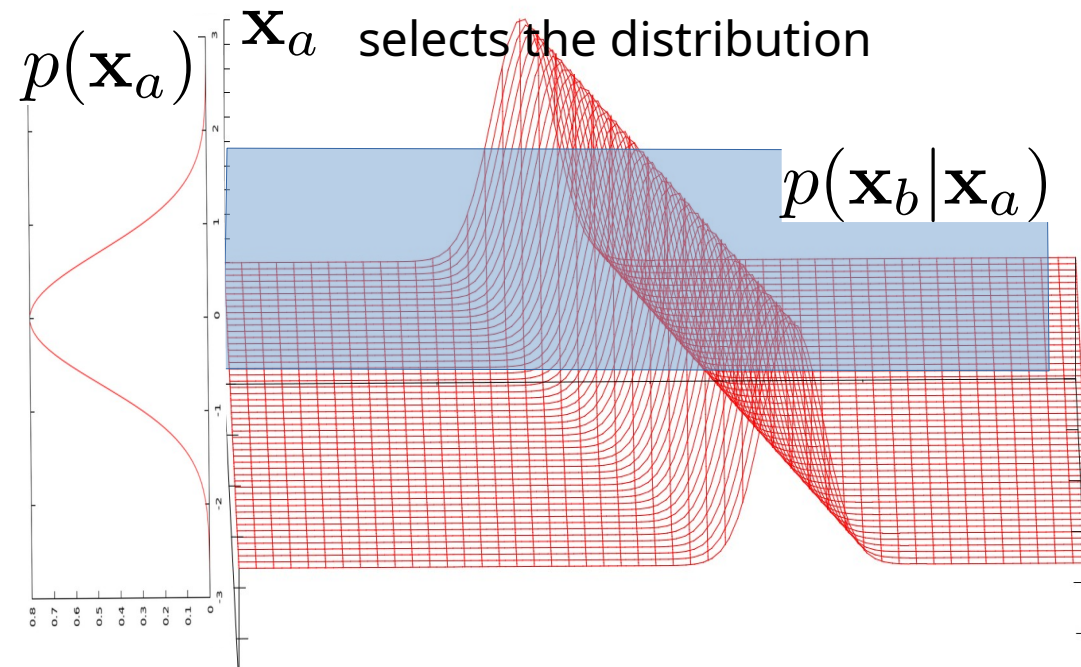
We want to compute

$$p(\mathbf{x}_a, \mathbf{x}_b) = \mathcal{N}(\mathbf{x}_{a,b}; \mu_{a,b}, \Sigma_{a,b})$$

The parameters are

$$\mu_{a,b} = \begin{pmatrix} \mu_a \\ \mu_b \end{pmatrix} = \begin{pmatrix} \mu_a \\ \mathbf{A}\mu_a + \mathbf{c} \end{pmatrix}$$

$$\Sigma_{a,b} = \begin{pmatrix} \mathbf{A}^T \Sigma_{b|a} \mathbf{A} + \Sigma_a & -\mathbf{A}^T \Sigma_{b|a} \\ -\Sigma_{b|a} \mathbf{A}^T & \Sigma_{b|a} \end{pmatrix}$$



References

Further details are here (warmly recommended):

- *Thomas Schoen*, On Manipulating the Multivariate Gaussian Density